

Chemical Equation Examples

This template provides a quick reference for typesetting chemical equations in LaTeX using the mhchem package which has been specifically designed to make chemical typesetting intuitive.

The template starts with simple chemical notation and progresses to complex chemical equations involving a variety of interactions and notations.

Description

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Basics H_2O , Sb_2O_3 , H^+ , CrO_4^{2-} , AgCl_2^- , $[\text{AgCl}_2]^-$, Y^{99+} , $\text{H}_{2(\text{aq})}$, NO_3^- , $(\text{NH}_4)_2\text{S}$

Amounts $2\text{H}_2\text{O}$, $\frac{1}{2}\text{H}_2\text{O}$

Isotopes $^{227}_{90}\text{Th}^+$

Special Symbols $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, $[\text{Cd}\{\text{SC}(\text{NH}_2)_2\}_2] \cdot [\text{Cr}(\text{SCN})_4(\text{NH}_3)_2]_2$, RNO_2^- , RNO_2^- , $\mu\text{-Cl}$

Bonds $\text{C}_6\text{H}_5\text{-CHO}$, A-B=C=D , A-B=C , $\text{A}\equiv\text{B}\equiv\text{C}\equiv\text{D}$, $\text{A}\cdots\text{B}\cdots\text{C}$, $\text{A}\rightarrow\text{B}\leftarrow\text{C}$

Using Math $\text{Fe}(\text{CN})_3$, $x\text{Na}(\text{NH}_4)\text{HPO}_4 \xrightarrow{\Delta} (\text{NaPO}_3)_x + x\text{NH}_3\uparrow + x\text{H}_2\text{O}$

Reaction Arrows $\text{CO}_2 + \text{C} \longrightarrow 2\text{CO}$, $\text{CO}_2 + \text{C} \longleftarrow 2\text{CO}$, $\text{CO}_2 + \text{C} \rightleftharpoons 2\text{CO}$, $\text{H}^+ + \text{OH}^- \rightleftharpoons \text{H}_2\text{O}$, $\text{A} \longleftrightarrow \text{A}$, $\text{CO}_2 + \text{C} \xrightarrow{\alpha} 2\text{CO}$, $\text{CO}_2 + \text{C} \xrightarrow[\beta]{\alpha} 2\text{CO}$, $\text{CO}_2 + \text{C} \xrightarrow[\text{below}]{\text{above}} 2\text{CO}$, $\text{A} \xrightarrow{+\text{H}_2\text{O}} \text{B}$

Precipitate and Gas $\text{SO}_4^{2-} + \text{Ba}^{2+} \longrightarrow \text{BaSO}_4\downarrow$, $\text{Zn} + \text{H}_2\text{SO}_4 \longrightarrow \text{ZnSO}_4 + \text{H}_2\uparrow$

Extra Examples $\text{Zn}^{2+} \xrightleftharpoons[+2\text{H}^+]{+2\text{OH}^-} \text{Zn}(\text{OH})_2\downarrow \xrightleftharpoons[+2\text{H}^+]{+2\text{OH}^-} [\text{Zn}(\text{OH})_4]^{2-}$, $K = \frac{[\text{Hg}^{2+}][\text{Hg}]}{[\text{Hg}_2^{2+}]}$, $\text{Hg}^{2+} \xrightarrow[\text{red}]{\text{I}^-} \text{HgI}_2 \xrightarrow[\text{red}]{\text{I}^-} [\text{HgI}_4]^{2-}$

Authorship

Current Version

This template was created by Vel using example chemical equations in the mhchem bundle documentation by Martin Hensel.

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